

CS4084

Quantum Computing

Week 03 – Lecture 06

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Quick Review: What We Covered Last Lecture

- **Dirac Notation (Part II):** Bras, kets, inner/outer products, and operators with physical interpretation.
- **Deterministic Operations:** Operator-based state evolution with fixed, reversible mappings.
- **Probabilistic Operations:** Measurement as a probabilistic process with state update after outcomes.
- **Composing Operations:** Sequential operators combined via matrix multiplication; order matters.

Key Message: Dirac Notation (second part) unifies deterministic evolution, probabilistic measurement, and composed quantum operations.

Dirac Notation (Third Part)

The Dirac notation can be used for arbitrary vectors: any name can be used in place of a classical state. Kets are column vectors, bras are row vectors.

Example

The notation $|\psi\rangle$ is commonly used to refer to an arbitrary vector:

$$|\psi\rangle = \frac{1+2i}{3} |0\rangle - \frac{2}{3} |1\rangle$$

$$\langle\psi| = \frac{1-2i}{3} \langle 0| - \frac{2}{3} \langle 1|$$

For any column vector $|\psi\rangle$, the row vector $\langle\psi|$ is the **conjugate transpose** of $|\psi\rangle$:

$$\langle\psi| = |\psi\rangle^\dagger$$

Example

Example

The notation $|\psi\rangle$ is commonly used to refer to an arbitrary vector:

$$|\psi\rangle = \frac{1+2i}{3} |0\rangle - \frac{2}{3} |1\rangle = \begin{pmatrix} \frac{1+2i}{3} \\ -\frac{2}{3} \end{pmatrix}$$

$$\langle\psi| = \frac{1-2i}{3} \langle 0| - \frac{2}{3} \langle 1| = \left(\frac{1-2i}{3} \quad -\frac{2}{3} \right)$$

Measuring quantum states

For this lesson will restrict our attention to *standard basis measurements*:

- The possible *outcomes* are the *classical states*.
- The probability for each classical state to be the outcome is the *absolute value squared* of the corresponding quantum state vector entry.

Example 1

Measuring the quantum state

$$|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle$$

yields an outcome as follows:

$$\Pr(\text{outcome is } 0) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad \Pr(\text{outcome is } 1) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

Example..

Example 2

Measuring the quantum state

$$|-\rangle = \frac{1}{\sqrt{2}}|0\rangle - \frac{1}{\sqrt{2}}|1\rangle$$

yields an outcome as follows:

$$\Pr(\text{outcome is } 0) = \left| \frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2} \quad \Pr(\text{outcome is } 1) = \left| -\frac{1}{\sqrt{2}} \right|^2 = \frac{1}{2}$$

Example..

Example 3

Measuring the quantum state

$$\frac{1+2i}{3} |0\rangle - \frac{2}{3} |1\rangle$$

yields an outcome as follows:

$$\Pr(\text{outcome is } 0) = \left| \frac{1+2i}{3} \right|^2 = \frac{5}{9} \quad \Pr(\text{outcome is } 1) = \left| -\frac{2}{3} \right|^2 = \frac{4}{9}$$

Unitary operations

The set of allowable **operations** that can be performed on a quantum state is different than it is for classical information.

Operations on quantum state vectors are represented by **unitary matrices**.

Definition

A square matrix $\mathbf{U}$ having complex number entries is **unitary** if it satisfies the equalities

$$\mathbf{U}^\dagger \mathbf{U} = \mathbb{1} = \mathbf{U} \mathbf{U}^\dagger$$

where $\mathbf{U}^\dagger$ is the conjugate transpose of $\mathbf{U}$ and $\mathbb{1}$ is the identity matrix.

Both equalities are equivalent to $\mathbf{U}^{-1} = \mathbf{U}^\dagger$.

Condition!

The condition that an $n \times n$ matrix U is unitary is equivalent to

$$\|Uv\| = \|v\|$$

for every n -dimensional column vector v with complex number entries.

If v is a quantum state vector, then Uv is also a quantum state vector.

Qubit Unitary operations

1. Pauli operations

Pauli operations are ones represented by the Pauli matrices:

$$\mathbb{1} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix} \quad \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \quad \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix} \quad \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$$

Common alternative notations: $X = \sigma_x$, $Y = \sigma_y$, and $Z = \sigma_z$.

The operation σ_x is also called a **bit flip** (or a NOT operation) and the σ_z operation is called a **phase flip**:

$$\sigma_x |0\rangle = |1\rangle$$

$$\sigma_z |0\rangle = |0\rangle$$

$$\sigma_x |1\rangle = |0\rangle$$

$$\sigma_z |1\rangle = -|1\rangle$$

Qubit Unitary operations

2. Hadamard operation

The Hadamard operation is represented by this matrix:

$$H = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}$$

Checking that H is unitary is a straightforward calculation:

$$\begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix}^\dagger \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1}{2} + \frac{1}{2} & \frac{1}{2} - \frac{1}{2} \\ \frac{1}{2} - \frac{1}{2} & \frac{1}{2} + \frac{1}{2} \end{pmatrix} = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}$$

Qubit Unitary operations

3. Phase operations

A phase operation is one described by the matrix

$$P_{\theta} = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\theta} \end{pmatrix}$$

for any choice of a real number θ .

The operations

$$S = P_{\pi/2} = \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \quad \text{and} \quad T = P_{\pi/4} = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$$

are important examples.

Example 1

$$H|0\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} \end{pmatrix} = |+\rangle$$

$$H|1\rangle = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} \frac{1}{\sqrt{2}} \\ -\frac{1}{\sqrt{2}} \end{pmatrix} = |-\rangle$$

Example 2

$$H|0\rangle = |+\rangle \quad H|+\rangle = |0\rangle$$

$$H|1\rangle = |-\rangle \quad H|-\rangle = |1\rangle$$

$$\begin{aligned} H\left(\frac{1+2i}{3}|0\rangle - \frac{2}{3}|1\rangle\right) &= \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} \frac{1+2i}{3} \\ -\frac{2}{3} \end{pmatrix} = \begin{pmatrix} \frac{-1+2i}{3\sqrt{2}} \\ \frac{3+2i}{3\sqrt{2}} \end{pmatrix} \\ &= \frac{-1+2i}{3\sqrt{2}}|0\rangle + \frac{3+2i}{3\sqrt{2}}|1\rangle \end{aligned}$$

Example 3

$$T|0\rangle = |0\rangle \quad \text{and} \quad T|1\rangle = \frac{1+i}{\sqrt{2}}|1\rangle$$

$$\begin{aligned} T|+\rangle &= T\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1}{\sqrt{2}}|1\rangle\right) \\ &= \frac{1}{\sqrt{2}}T|0\rangle + \frac{1}{\sqrt{2}}T|1\rangle \\ &= \frac{1}{\sqrt{2}}|0\rangle + \frac{1+i}{2}|1\rangle \end{aligned}$$

$$T = \begin{pmatrix} 1 & 0 \\ 0 & \frac{1+i}{\sqrt{2}} \end{pmatrix}$$

Example 4

$$T|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1+i}{2}|1\rangle$$

$$\begin{aligned} HT|+\rangle &= H\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1+i}{2}|1\rangle\right) \\ &= \frac{1}{\sqrt{2}}H|0\rangle + \frac{1+i}{2}H|1\rangle \\ &= \frac{1}{\sqrt{2}}|+\rangle + \frac{1+i}{2}|-\rangle \end{aligned}$$

$$H|0\rangle = |+\rangle$$

$$H|1\rangle = |-\rangle$$

Final Answer!

$$T|+\rangle = \frac{1}{\sqrt{2}}|0\rangle + \frac{1+i}{2}|1\rangle$$

$$\begin{aligned} HT|+\rangle &= H\left(\frac{1}{\sqrt{2}}|0\rangle + \frac{1+i}{2}|1\rangle\right) \\ &= \frac{1}{\sqrt{2}}H|0\rangle + \frac{1+i}{2}H|1\rangle \\ &= \frac{1}{\sqrt{2}}|+\rangle + \frac{1+i}{2}|-\rangle \\ &= \left(\frac{1}{2}|0\rangle + \frac{1}{2}|1\rangle\right) + \left(\frac{1+i}{2\sqrt{2}}|0\rangle - \frac{1+i}{2\sqrt{2}}|1\rangle\right) \\ &= \left(\frac{1}{2} + \frac{1+i}{2\sqrt{2}}\right)|0\rangle + \left(\frac{1}{2} - \frac{1+i}{2\sqrt{2}}\right)|1\rangle \end{aligned}$$

Composing Unitary Operations

Compositions of unitary operations are represented by **matrix multiplication** (similar to the probabilistic setting).

Example: square root of NOT

Applying a Hadamard operation, followed by the phase operation S , followed by another Hadamard operation yields this operation:

$$HSH = \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} \begin{pmatrix} 1 & 0 \\ 0 & i \end{pmatrix} \begin{pmatrix} \frac{1}{\sqrt{2}} & \frac{1}{\sqrt{2}} \\ \frac{1}{\sqrt{2}} & -\frac{1}{\sqrt{2}} \end{pmatrix} = \begin{pmatrix} \frac{1+i}{2} & \frac{1-i}{2} \\ \frac{1-i}{2} & \frac{1+i}{2} \end{pmatrix}$$

Applying this unitary operation twice yields a NOT operation:

$$(HSH)^2 = \begin{pmatrix} \frac{1+i}{2} & \frac{1-i}{2} \\ \frac{1-i}{2} & \frac{1+i}{2} \end{pmatrix}^2 = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix}$$

Summary: Key Takeaways

- **Quantum Information:** States, measurements, and evolution are described using complex vector spaces.
- **Dirac Notation (Part III):** Operators and projectors act on states to model dynamics and measurement.
- **Quantum Measurement:** Outcomes are probabilistic; probabilities are given by squared amplitudes, and measurement changes the state.
- **Unitary Operations:** Quantum evolution is reversible, probability-preserving, and represented by unitary matrices.
- **Qubit Unitaries:** Single-qubit gates are 2×2 unitary operators.
- **Composing Unitaries:** Sequential operations are combined via matrix multiplication, and order matters.

Next: Observables and Operators, Eigenvalues, Eigenvectors, Tensor products, Postulates of quantum mechanics, Introduction to Superposition